

THERMODYNAMICS
ChE 322 – Spring 2025
Unique # - 16025

Class Meeting: CPE 2.216 on TTh from 11:00 am – 12:30 pm
Instructor: Prof. Gabriel E. Sanoja (he/him), EER 5.618, gesanoja@che.utexas.edu

Introductory course in thermodynamics with special reference to chemical processes: basic laws, thermodynamic properties of single component systems, expansion and compression of fluids, heat engines, multicomponent systems, physical equilibrium, and chemical equilibrium.

TEXTBOOKS: **Required:** Koretsky, M. D. *Engineering and Chemical Thermodynamics*. John Wiley & Sons, Inc., 2012.
Optional: Sandler, S. I. *Chemical, Biochemical, and Engineering Thermodynamics*. 5th Ed. John Wiley & Sons, Inc., 2017.

PREREQUISITES: Chemical Engineering 210 and 317, Chemistry 353, and Mathematics 427J with a grade of at least C- in each.

TEACHING ASSISTANT: Devin Rosmarin, devin.rosmarin@utmail.utexas.edu

TUTOR: Amogh Mahambare, amoghmahambare@utexas.edu

RECITATION: W 1:00 - 3:00 p.m. BUR 108

OFFICE HOURS: MT 5:00 – 6:00 pm Sanoja – EER 5.702
M 2:00 – 3:00 pm Rosmarin – EER 5.704
T 2:00 – 3:00 pm Rosmarin – EER 5.704

TUTORING: M 3:00 – 5:00 pm Mahambare – EER 5.704
F 3:00 – 5:00 pm Mahambare – EER 5.702

PARTICIPATION: There will be an assessment of your engagement and participation in the course which will represent 5% of your letter grade. **You are expected to attend lecture, recitation, and some office hours; arrive on time; turn off and stow away electronic devices; and ask questions. Another way to actively participate in the course is to submit a detailed solution to one (1) homework problem, either in LaTeX or Microsoft Word, through Canvas by the assigned deadline.**

QUIZZES: **There will be eleven (11) quizzes administered throughout the semester at the beginning of recitation.** I will disregard your lowest quiz grade but please be aware that missed quizzes will receive no credit and cannot be made up. Quizzes will account for 20% of your grade.

HOMEWORKS: There will be homework nearly every week. These will be composed of traditional mathematically-oriented problems, but also of some readings and videos related to soft-skills and history of engineering sciences. Homework will not be collected nor graded, but it will be the basis of the quizzes. **You are expected to actively learn by doing homework so that you can score perfectly on the quizzes. Also, you are encouraged to submit a detailed solution to one (1) homework problem, either in LaTeX or Microsoft Word, through Canvas by the Sunday at 5 pm to gain participation credit in the course.**

MIDTERMS AND FINAL: **There will be two (2) midterms administered during recitation, and one (1) final administered during final week.** These will be closed book and notes, but

allow you one (1) US Letter size-paper with content of your choice as supportive information. Please be aware that missed exams cannot be made up. If you miss one (1) midterm due to very exceptional circumstances, you will receive the weighted-grade of the final. If you miss both midterms, you will be referred to the Dean's Office (ECJ 2.200) and subjected to administrative proceedings. If you do not miss any midterm, your lowest midterm-grade will be substituted with that of the weighted final-grade in as long as this is beneficial for your overall performance in the course. Each midterm will account for 20% of your grade, and the final exam for 35%.

If you want to request an exam regrade, please write an explanatory note, attach it to your exam, and submit it directly to me within one class period. Failure to comply with this procedure could result in denial of your request. Please be aware that I will regrade the entire exam.

EVALUATION:

Quizzes*	20% - 2% each
Midterms [#]	40% - 20% each
Participation	5%
Final	35%

*Lowest grades will be disregarded. Missed quizzes will receive no credit and count towards the lowest grades.

[#]You may substitute the lowest grade of a Midterm with that of the Final, properly weighted, as long as (i) you do not miss any Midterm and (ii) this substitution is beneficial for your letter grade.

Letter grades will be determined using the following scale:

A	A ⁻	B ⁺	B	B ⁻	C ⁺	C	C ⁻	D ⁺	D	D ⁻	F
100-90	89-85	84-80	79-75	74-70	69-66	65-63	62-60	59-56	55-53	52-50	49-0

Please be cautious when using the average calculated by Canvas to determine your standing in the course. This average is calculated automatically and might be inaccurate in some instances (e.g. it does not account for the disregarded quiz grade).

If you fail to score nearly perfectly on the first two quizzes, you are strongly encouraged to drop the course. **You are expected to actively study from the very beginning of the semester to perform well. This type of study entails attending lecture, recitation, and office hours; as well as critically doing the homework.** The instructor, TA, and tutor are available in Office Hours and Tutoring to clarify concepts and help you master problem-solving skills, but you are ultimately responsible for learning how to solve problems by yourself.

ABET OUTCOME:

An ABET (Accreditation Board for Engineering and Technology) outcome of the course will be for students to have the **“ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.”** AIChE (American Institute for Chemical Engineers) defines a complex engineering problem as having one or more of the following characteristics: **“involving wide-ranging or conflicting technical issues, having no obvious solution, addressing problems not encompassed by current standards and codes, involving diverse groups of stakeholders, including**

many component parts or sub-problems, involving multiple disciplines, or having significant consequences in a range of contexts.”

ACCOMMODATIONS:

The University of Texas at Austin is committed to create an accessible and inclusive learning environment consistent with university policy, and federal and state law. **If you experience any learning barriers, please let me know so that I can work to ensure you have equal opportunity to participate and succeed in the course.** If you are a student with a disability and need accommodation, please contact [Services for Students with Disabilities \(SSD\)](#). If you are already registered with SSD, please submit your Accommodation Letter directly to me as early in the semester as possible.

PRONOUNS:

Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with differences of race, culture, religion, politics, sexual orientation, gender, gender variance, and nationalities. Class rosters are provided to the instructor with the student’s legal name, unless they have added a “preferred name” with the [Gender and Sexuality Center](#). I will gladly honor your request to address you by a name that is different from what appears on the official roster, and by the gender pronouns you use (she/he/they/ze, etc). Please advise me of any changes early in the semester so that I can update my records. You are also welcome to [add your pronouns to Canvas](#).

ACADEMIC INTEGRITY:

Each student in the course is expected to abide by The University of Texas Honor Code: “As a student at The University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity”. **You are responsible for understanding [UT’s Academic Honesty and the University Honor Code](#).**

TITLE IX REPORTING:

Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence and stalking at federally funded educational institutions. UT Austin is committed to fostering a learning and working environment free from discrimination in all its forms. When sexual misconduct occurs in our community, the university can:

1. Intervene to prevent harmful behavior from continuing or escalating.
2. Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation.
3. Investigate and discipline violations of the university’s relevant policies.

Beginning January 1, 2020, Texas Senate Bill 212 requires all employees of Texas universities, including faculty, to report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. **I am a Responsible Employee and must report any Title IX related incidents that are disclosed in writing, discussion, or one-on-one. Before talking with me, or with any faculty or staff member about a Title IX related incident, be sure to ask whether they are a responsible employee.** If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, contact the [Title IX Office](#) via email at titleix@austin.utexas.edu, or call 512-471-0419.

Although graduate teaching and research assistants are not subject to Texas Senate Bill 212, they are still mandatory reporters under Federal Title IX laws and are required to report a wide range of behaviors we refer to as sexual misconduct, including the types of sexual misconduct covered under Texas Senate Bill 212. The Title IX office has developed supportive ways to respond to a survivor and compiled campus resources to support survivors.

IMPORTANT DATES (See also [Academic Calendar 2024-2025](#))

Jan. 14 – T.	First class in CPE 2.216 from 11:00 am to 12:30 pm
Jan. 15 – W.	First recitation in BUR 108 from 1:00 pm to 3:00 pm.
Jan. 16 – Th.	Last day to add a class without permission.
Jan. 20 – M.	Martin Luther King Jr. Day; no Office Hours held.
Jan. 22 – W.	Quiz #1 in BUR 108 from 1:00 pm to 1:30 pm.
Jan. 29 – W.	Quiz #2. Last day to drop a class without permission.
Feb. 05 – W.	Quiz #3.
Feb. 12 – W.	Quiz #4.
Feb. 18 – T.	Prof. Sanoja on travel; no Lecture held.
Feb. 19 – W.	Quiz #5.
Feb. 21 – F.	Make-up lecture from 3:00 pm to 4:30 pm; room TBD.
Feb. 26 – W.	Midterm #1 in BUR 108 from 1:00 pm to 3:00 pm.
Mar. 05 – W.	Quiz #6.
Mar. 12 – W.	Quiz #7.
Mar. 17 – 22.	Spring Break.
Mar. 25 – T.	Prof. Sanoja on travel; no Lecture held.
Mar. 26 – W.	Quiz #8.
Mar. 28 – F.	Make-up lecture from 3:00 pm to 4:30 pm; room TBD.
Apr. 02 – W.	Quiz #9.
Apr. 09 – W.	Quiz #10.
Apr. 15 – T.	Prof. Sanoja on travel; no Lecture held.
Apr. 16 – W.	Quiz #11. Last day an undergraduate may: Q-drop a class; withdraw; change a class to pass/fail.
Apr. 18 – F.	Make-up lecture from 3:00 pm to 4:30 pm; room TBD.
Apr. 23 – W.	Midterm #2 in BUR 108 from 1:00 pm to 3:00 pm.
Apr. 28 – M.	Last class day. Last day an undergraduate may, with the required approvals, request a non-academic Q-drop. Last day an undergraduate may register in absentia.
May. 01 – 05.	Final Exam at time and place TBD.

This syllabus is from January 12th, 2025 and is subject to change. Students who miss class are responsible for learning about any changes announced in their absence.

COURSE OUTLINE:

The course is composed of three (3) modules and twenty-eight (28) lectures; some of which should be supplemented with book readings.

Module 1: Basic Laws of Thermodynamics

1. Course Overview.
2. Introductory Concepts. **K 2.1-2.3**
3. The First Law of Thermodynamics. Closed Systems. **K 2.4.**
4. Open Systems. **K 2.5, 2.8.**
5. Thermodynamic Cycles and the Carnot Cycle **K 2.9**
6. The Second Law of Thermodynamics. **K 3.1-3.3.**
7. Closed and Open Systems. **K. 3.5-3.6.**
8. Introduction to Rankine Cycle. **K 3.8-3.9.**
9. Rankine Cycle. **K 3.9.**

Module 2: Intermolecular Forces, Equations of State, and Thermodynamic Web

10. Intermolecular Forces. **K 4.1-4.2.**
11. Equations of State for Single-Component Systems. **K 4.3.**
12. van der Waals Equation and Mixtures. **K 4.3-4.4.**
13. Thermodynamic Relationships. **K 5.1-5.3.**
14. Departure Functions.

Module 3: Phase Equilibria

15. Phase Equilibrium for Single-Component Systems. **K 6.1-6.2.**
16. Thermodynamics of Mixtures. **K 6.3.**
17. Multicomponent Phase Equilibrium. **K 6.4.**
18. Fugacity in the Vapor Phase. **K. 7.1-7.3.**

19. Fugacity of Vapor Mixtures. **K 7.3-7.4.**
20. Fugacity in the Liquid Phase. **K 7.4.**
21. Excess Gibbs Energy and Activity Coefficient Models. **K 7.4.**
22. Vapor-Liquid Equilibrium (VLE). **K 8.1.**
23. Azeotropes and Gas Solubilities in Liquids. **K 8.1.**
24. Liquid-Liquid Equilibrium (LLE). **K 8.2.**
25. Colligative Properties. **K 8.5.**

Module 4: Chemical Reaction Equilibrium.

26. Chemical Reaction and Gibbs Energy. **K 9.1-9.2.**
27. Equilibrium for a Single Reaction. **K 9.3.**
28. Calculation of K . **K 9.4.**